

STUDENT PERFORMANCE PREDICTION SYSTEM

PREDICT ● IDENTIFY RISK ● PROVIDE AI POWERED INTERVENTION

BIT4543 ARTIFICIAL INTELLIGENCE

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INTRODUCTION

Background

- Educational institutions collect large amounts of student data.
- Traditional monitoring depends on examinations, assessments and manual reviews.
- Students with academic difficulties may be identified too late.
- AI and Machine Learning can analyse data, find patterns and support early intervention.

Key Idea

Predict student outcomes → identify risks → provide appropriate intervention.



PROBLEM STATEMENT & RESEARCH GAP

Existing Problems

- Delayed Risk Identification
- Lack of Predictive Insights
- Ineffective Intervention Alignment

Research Gap

Prediction alone is not enough.

A prediction does not automatically tell us what factor is causing the risk, what action to take, or how to communicate the recommendation.

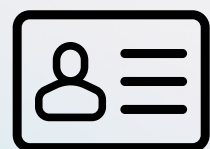


DATASET : OULAD



32,593 students

22 course presentations



80% Training (26,074 students)

20% Testing (6,519 students)

Open University Learning Analytics Dataset
Provides anonymised data including:

- Student information
- Registration information
- Assessment results
- Course/module information
- Assessment information
- VLE activities



DATA PREPROCESSING & FEATURES



Load Data



Inspect Data



Feature Selection/
Engineering



Target Preparation



Train/Test Split

Key Features (10 total)

- Average assessment score
- Assessment count
- Studied credits
- Number of previous attempts
- + 6 other features (categorical and numerical)

Target Classes

- Distinction
- Pass
- Fail
- Withdrawn



PROJECT OBJECTIVES



Develop an AI-based Student Performance Prediction System



Prepare and process student data from CULAD



Develop and evaluate a Random Forest model



Develop a knowledge base and rule-based risk workflow



Integrate Generative AI for intervention recommendations



Develop an application prototype for user interaction



KNOWLEDGE BASE & RISK DETECTION

Risk Levels

Low → Medium → High

Risk Factors & Rules

RISK FACTOR	THRESHOLD
LOW GPA	GPA < 2.0
POOR ATTENDANCE	ATTENDANCE < 75%
MISSED ASSIGNMENTS	23
LOW PARTICIPATION	< 5/10



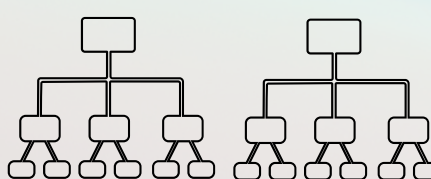
MACHINE LEARNING MODEL

Random Forest Classifier

Why Random Forest?

- Suitable for classification
- Works well with tabular data
- Models non-linear relationships
- Combines multiple decision trees
- Provides feature importance

Multiple Declaration



Multiple Decision Trees

Final Prediction



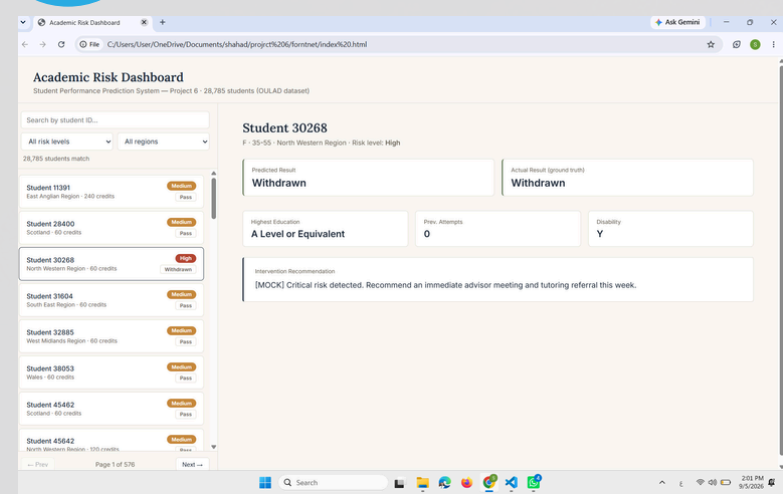
AI WORKFLOW & GEMINI INTEGRATION

How it works?

- Student Risk Profile
- Identity Risk Factors
- Retrieve Risk-Level Guidance
- Retrieve Risk-Factor Guidance
- Build Knowledge Context
- Add Student Information
- Apply Prompt Engineering
- Send to Gemini API
- Generate intervention Recommendation



APPLICATION



MODEL PERFORMANCE & EVALUATION

Overall Performance		Class Performance			
Metric	Result	Class	Precision	Recall	F1
Accuracy	44.62%	Distinction	0.36	0.01	0.01
Weighted Precision	0.43	Fail	0.41	0.06	0.11
Weighted Recall	0.45	Pass	0.44	0.73	0.55
Weighted F1-score	0.38	Withdrawn	0.46	0.49	0.48



TESTING AND AI WORKFLOW VALIDATION

Risk	GPA	Attendance	Missed Assignments	Participation
Low	3.4	91%	0	8/10
Medium	2.6	78%	2	5/10
High	1.9	60%	4	3/10

Limitation

- Moderate model accuracy
- Limited feature set
- Class imbalance
- OULAD represents a specific educational environment
- Prototype rather than production system
- Rule-based thresholds require validation
- Generative AI depends on prompt/context/API behaviour

Future Improvement

- Add VLE activity data
- Include assignment submission patterns
- Use historical academic performance
- Compare XGBoost, Gradient Boosting, SVM and Neural Networks
- Apply SMOTE/oversampling
- Introduce real-time university data
- Improve UI/dashboard
- Add database integration
- Conduct evaluation with students, lecturers and advisors

Conclusion

Successfully integrates:

Machine Learning + Rule-Based Reasoning + Knowledge Base + Prompt Engineering + Generative AI

From prediction alone to action-oriented academic intervention.

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